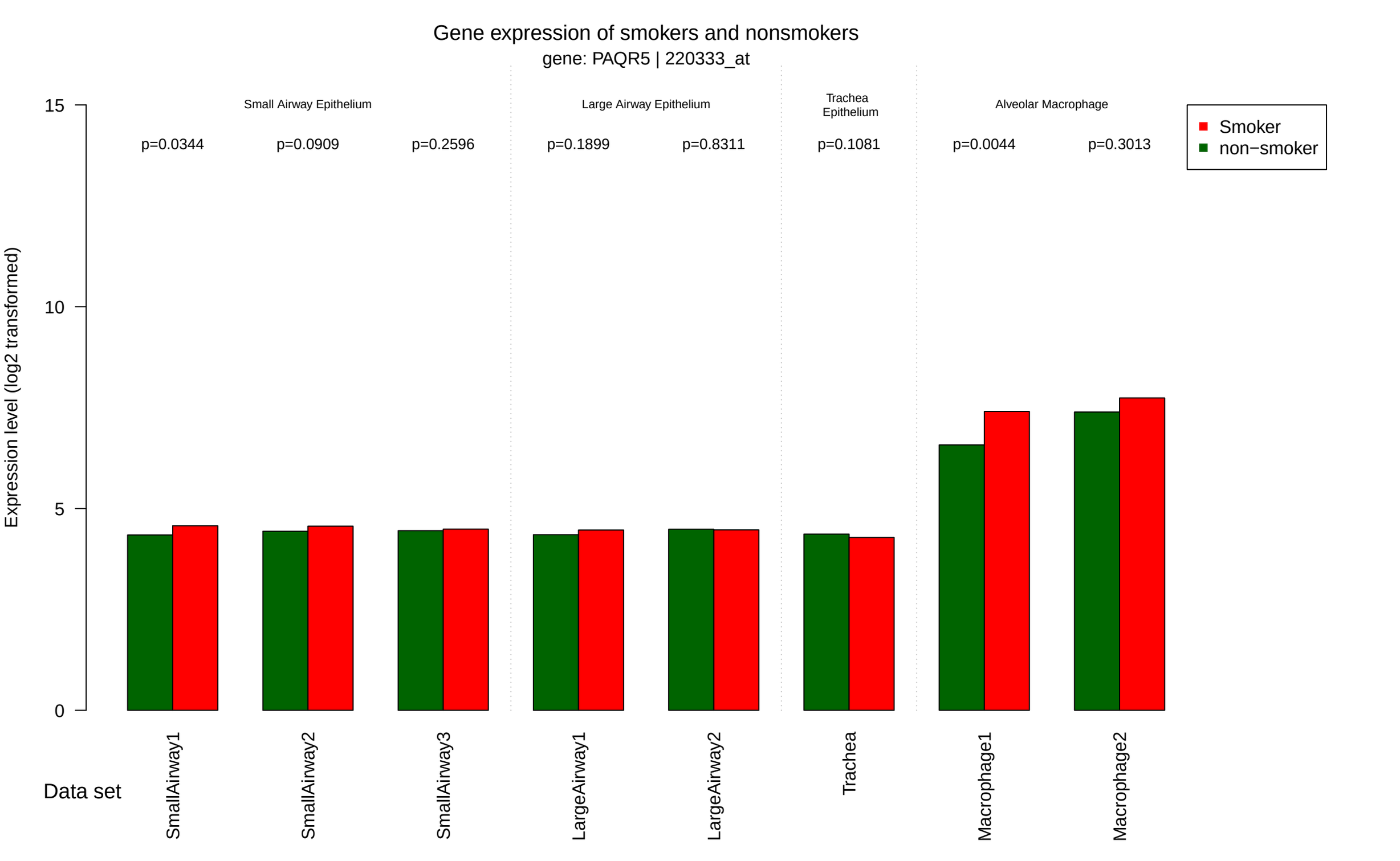
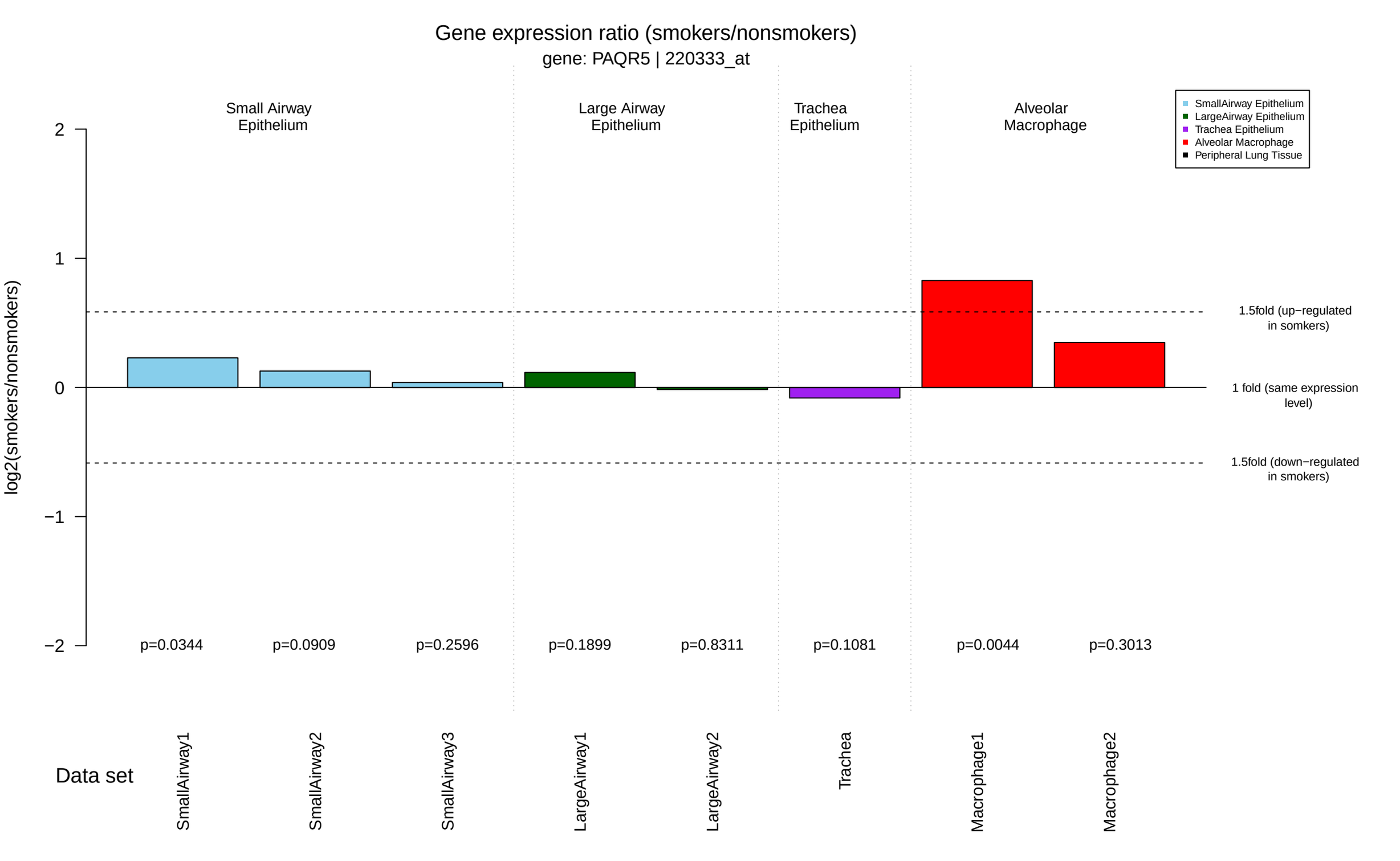
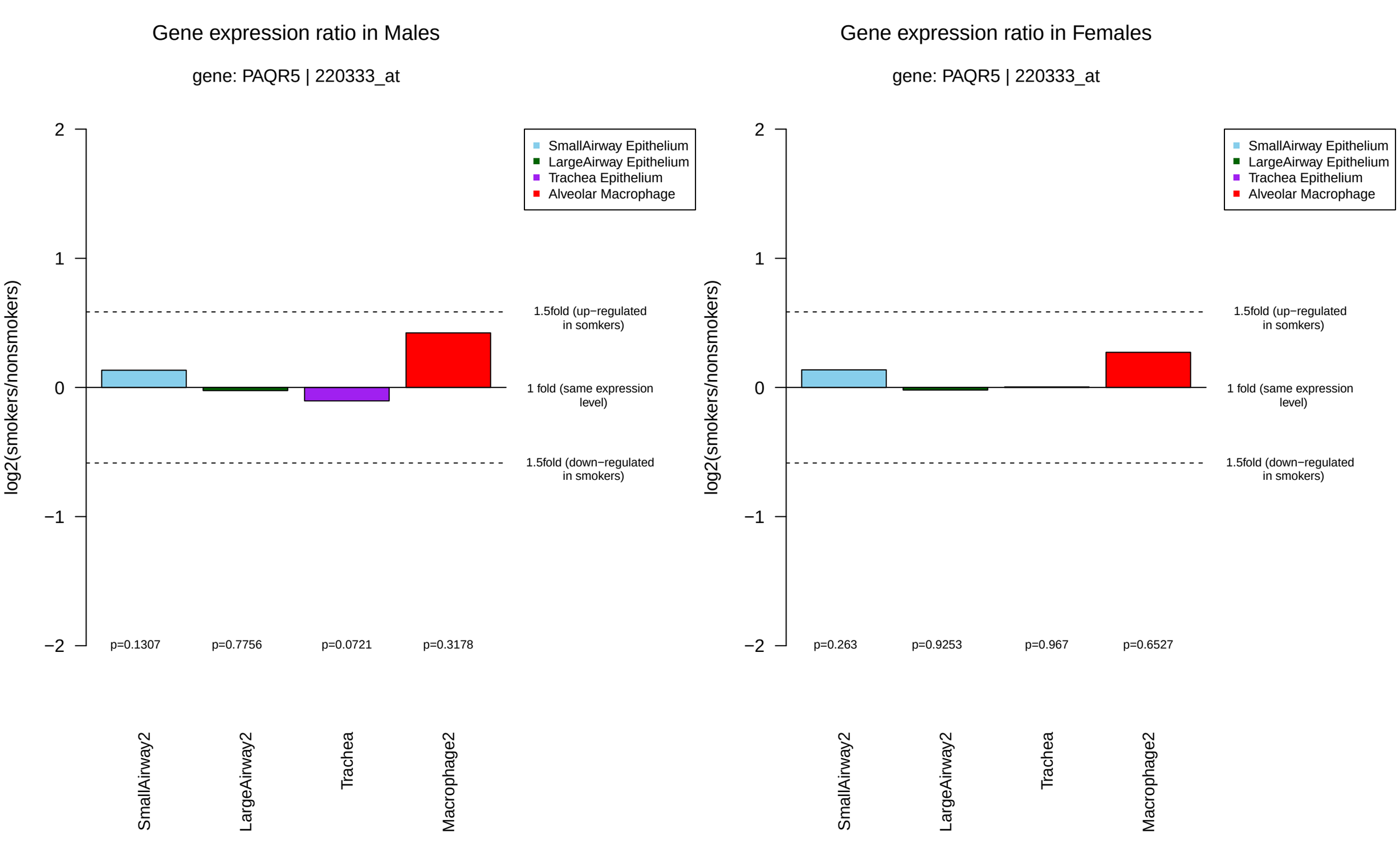
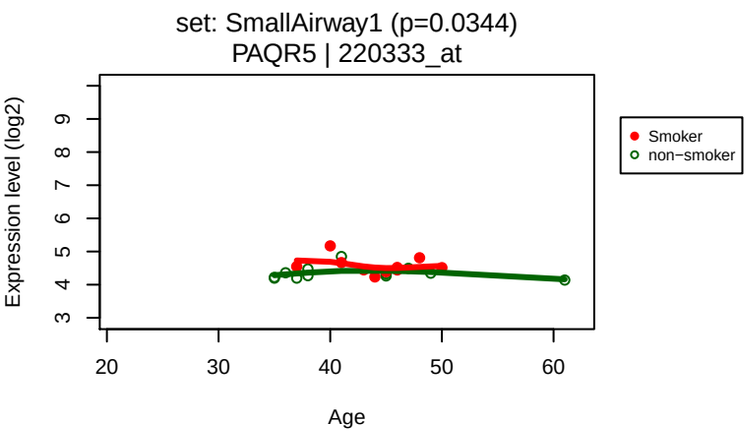
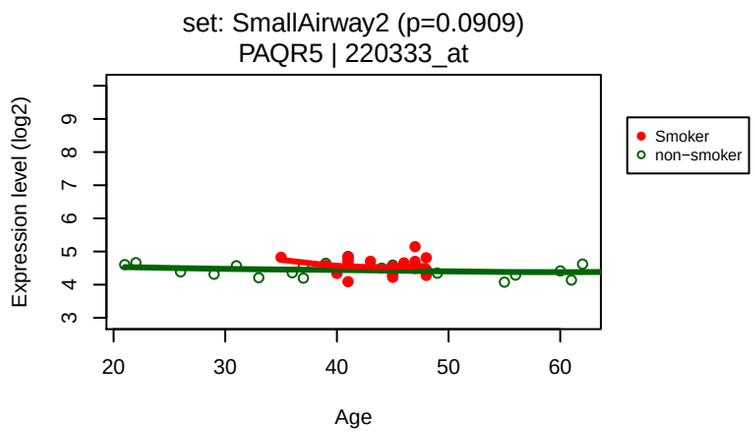
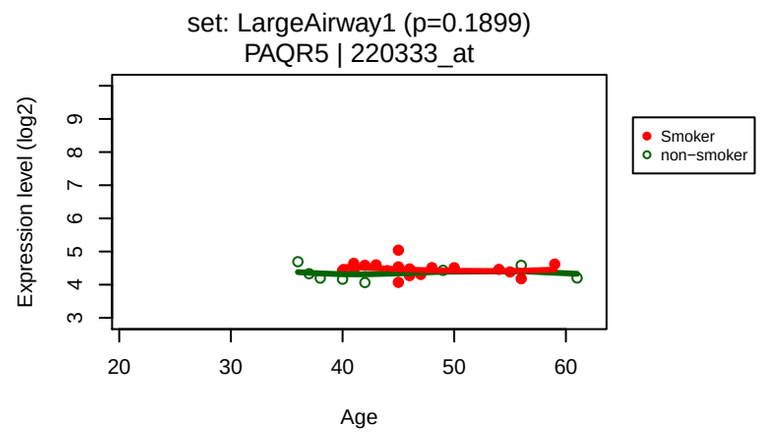

Gene expression of smokers and nonsmokers
gene: PAQR5 | 220333_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0344
p=0.0909
p=0.2596
p=0.1899
p=0.8311
p=0.1081
p=0.0044
p=0.3013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

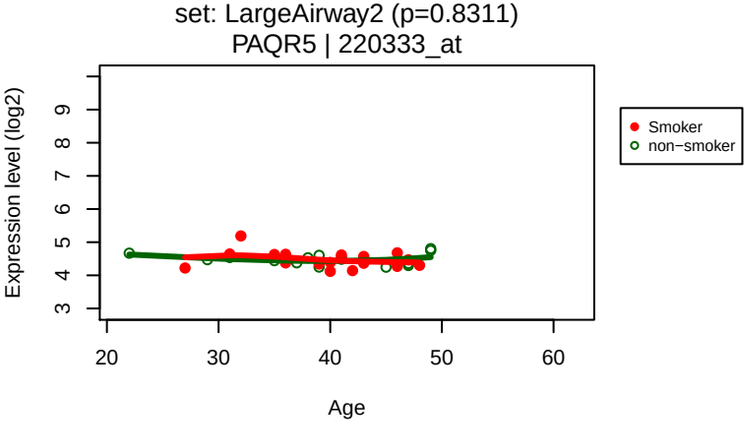

Gene expression ratio (smokers/nonsmokers)
gene: PAQR5 | 220333_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0344
p=0.0909
p=0.2596
p=0.1899
p=0.8311
p=0.1081
p=0.0044
p=0.3013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

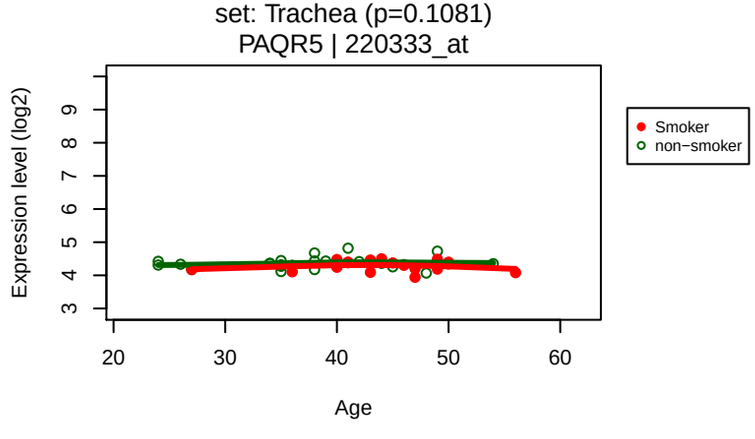

Gene expression ratio in Males
gene: PAQR5 | 220333_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1307
p=0.7756
p=0.0721
p=0.3178
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAQR5 | 220333_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.263
p=0.9253
p=0.967
p=0.6527
SmallAirway2
LargeAirway2
Trachea
Macrophage2

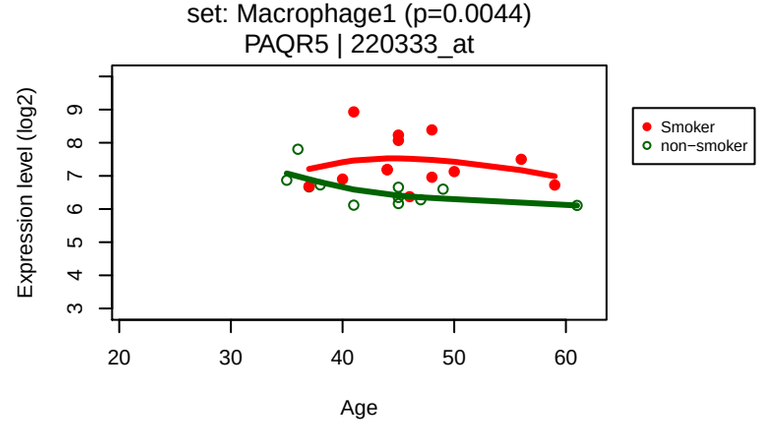

set: SmallAirway1 (p=0.0344)
PAQR5 | 220333_at
Expression level (log2)
Age
Smoker
non−smoker

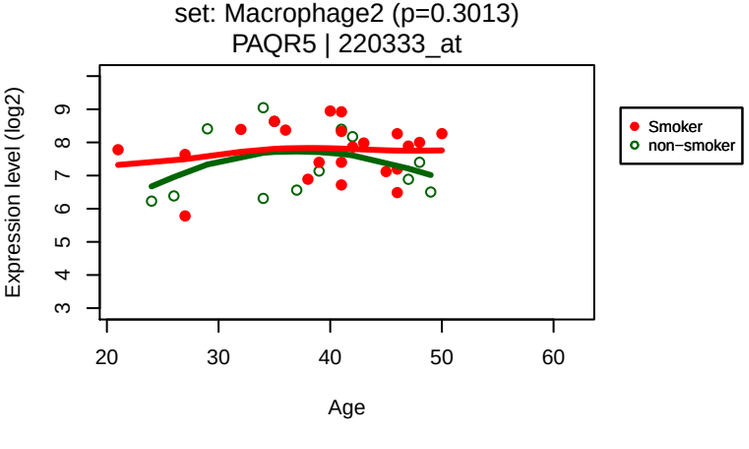

set: SmallAirway2 (p=0.0909)
PAQR5 | 220333_at
Expression level (log2)
Age
Smoker
non−smoker

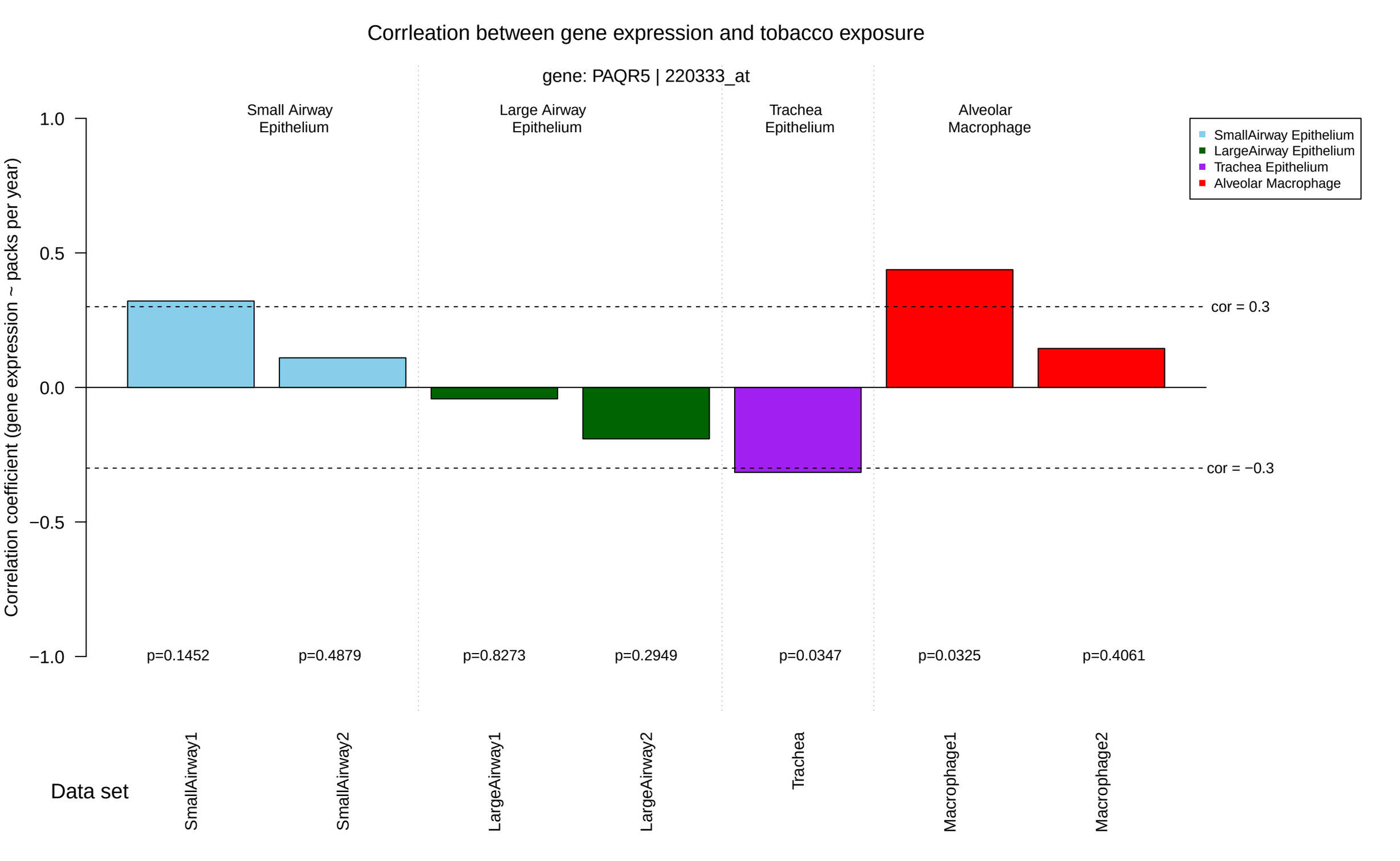

set: LargeAirway1 (p=0.1899)
PAQR5 | 220333_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8311)
PAQR5 | 220333_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1081)
PAQR5 | 220333_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0044)
PAQR5 | 220333_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3013)
PAQR5 | 220333_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAQR5 | 220333_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1452
p=0.4879
p=0.8273
p=0.2949
p=0.0347
p=0.0325
p=0.4061
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2